

Gamified Working Memory Intervention Enhances Prefrontal Neurocognitive Plasticity During Aging

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Considering the critical role of working memory (WM) in age-related cognitive decline, WM interventions are proposed as an effective approach for promoting healthy aging. However, traditional WM interventions often yield limited results, possibly due to a lack of interest and challenge. In light of the recent prominent trend of gamification, this study developed a challenging gamified WM intervention aimed at improving WM and providing extensive cognitive benefits for older adults, along with entertainment. Seventy-six older adults were randomly assigned to either the WM intervention group or the active control group. All participants underwent twelve 1-hr training sessions spanning 6 weeks. For each participant, cognitive assessments were completed at pretest, posttest, and 6 months after the intervention while structural magnetic resonance imaging (MRI) and resting-state functional MRI scans were conducted before and after intervention. The results revealed significant improvements in WM, inhibitory control, visuospatial processing, and episodic memory among older adults following the intervention. The left frontal pole, located in the rostral prefrontal cortex (rPFC), exhibited increased cortical thickness accompanied by reduced functional homogeneity and weakened connectivity with the left inferior temporal gyrus. Further statistical learning analysis demonstrated a consistent relationship between the changes at the left frontal pole and improved WM performance, suggesting that the rPFC may support WM and play a protective role in brain aging. The findings highlight the broad benefits of the gamified WM intervention on neurocognitive plasticity in the rPFC of older adults, indicating its potential as a promising tool for mitigating neurocognitive decline during the aging process.

Public Significance Statement

This study showcases the transformative potential of gamified working memory interventions in enhancing cognitive functions and promoting healthy aging in older adults. By targeting the rostral prefrontal cortex, this innovative approach not only improves working memory but also offers broader cognitive benefits, pointing toward a promising strategy to combat age-related neurocognitive decline.

Keywords: gamification, working memory intervention, the rostral prefrontal cortex, neurocognitive plasticity, older adults

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Population aging is the most important demographic problem worldwide (Aman et al., 2021). To mitigate the socioeconomic impacts of population aging and promote healthy aging, interventions that maintain the cognitive function of older adults are crucial. Research indicated that even delaying cognitive decline in older adults by 1 year could prevent nearly 10 million new cases of dementia (Ziegler et al., 2022).

Working memory (WM) plays a fundamental role in cognitive function, serving as a workspace for information (Bouchacourt & Buschman, 2019). Higher order cognitive abilities, such as reasoning, planning, and decision-making, rely heavily on this workspace to maintain, manipulate, and update information (Salmi et al., 2018). WM is closely associated with practical abilities necessary for daily activities, such as maintaining focused attention and following instructions (Coats et al., 2021). Impairments in WM significantly contribute to age-related cognitive decline and severe cognitive deterioration associated with neurodegenerative diseases like Alzheimer's disease (Reinhart & Nguyen, 2019). Consequently, the therapeutic potential of WM interventions for older adults is substantial, as such interventions have the ability to produce enduring enhancements in broad cognitive abilities, prevent neurodegenerative diseases, and sustain older adults' functional independence (Turnbull et al., 2022). However, compelling evidence reporting the effectiveness of WM interventions is relatively scarce. While WM interventions have been reported to have beneficial effects on cognitive and practical problems (Au et al., 2015; Lundqvist et al., 2010), these positive outcomes are outnumbered by studies reporting null or limited effects of WM intervention (Melby-Lervåg & Hulme, 2013; Soveri et al., 2017). Some studies have shown that WM intervention in older adults produces significant near-transfer effects on WM and far-transfer effects on other cognitive functions such as attention, inhibitory control, processing speed, and fluid intelligence (Borella et al., 2010; Brum et al., 2020). However, other studies have reported that WM intervention only improves performance on trained tasks without producing broader transfer effects (Booth et al., 2023; Sala et al., 2019; von Bastian et al., 2013). Regarding the long-term effects of WM interventions, although one meta-analysis demonstrated robust long-term near- and far-transfer effects (Hou et al., 2020), another meta-analysis did not show significant maintenance effects at follow-up (Teixeira-Santos et al., 2019).

Traditional WM interventions suffer from predictability, repeatability, and a lack of feedback and rewards, which can reduce participant motivation and engagement, hindering the effectiveness of the intervention (Blagden & De Rond, 2018; van de Weijer et al., 2019). Gamification, which entails integrating game design elements into nongame contexts (Hamari & Koivisto, 2015), offers a potential solution to address these limitations. Gamification incorporates elements such as point scoring, rewards, competition, feedback on performance, and the opportunity for self-discovery through play, making it inherently motivating and capable of significantly engaging participants in research and interventions (Long et al., 2023; Wang et al., 2024). Gamification, as an optimal approach that simultaneously fulfills utilitarian and hedonic needs (Koivisto & Hamari, 2019), is gaining popularity in diverse fields and is widely acknowledged as a promising nonpharmacological tool for enhancing cognitive functions in older adults (Vallejo et al., 2017).

There are a variety of commercial gamified cognitive training platforms that claim to enhance cognitive functions or prevent

cognitive decline in older adults. Popular platforms include Lumosity, Cogmed, Cognifit, Brain Age, and BrainHQ (Shah et al., 2017). Previous studies investigating these platforms have reached mixed conclusions about their effectiveness and the generalizability of training outcomes (McLaughlin, 2016). For example, one study using the adaptive WM game Cogmed found improvements in WM and attention in older adults (Brehmer et al., 2012), but another study did not find that WM was significantly enhanced after training (Flak et al., 2019). Training with Brain Age improved executive function and processing speed in older adults, but no transfer effects were observed on global cognition or attention (Nouchi et al., 2012). A meta-analysis examining the cognitive training effects of seven commercial online gamified platforms (BrainGymmer, BrainHQ, Cogmed, Cognifit, Dakim, Lumosity, and MyBrainTrainer) in older adults found that the near-transfer effects were small, and far-transfer effects were not significant in terms of objective measures of cognitive function (Nguyen et al., 2022). However, older adults who were trained with Brain HQ's eight cognitive games showed improvements in WM, inhibitory control, and processing speed (Meltzer et al., 2023). Although the gamification features of the commercially cognitive training platforms enhance user engagement and motivation, their training tasks primarily replicate traditional neuropsychological tasks that do not consistently promote new learning or the development of novel sequences of cognitive processes (Gathercole et al., 2019).

WM is a cognitive system that encompasses processes such as encoding, storage, rehearsal, retrieval, and manipulation of information (Allen et al., 2006; Wager & Smith, 2003). According to the cognitive routine framework, the limited effects of WM intervention arise from a failure to acquire a new cognitive routine—a novel, coordinated sequence of these processes developed within and extending beyond WM (Gathercole et al., 2019). The learning process follows three phases: cognitive, associative, and autonomous (Fitts & Posner, 1967). If a learned WM task can rely on existing processes that have already been fully developed and reached the autonomous stage, there will be limited opportunity for additional learning or refinement through intervention (Gathercole et al., 2019). Therefore, to achieve effective WM interventions, it is crucial to have a complex and challenging task that necessitates the development of new cognitive routines. During the initial phase, cognitive resources are required to manage and execute existing processes in a novel sequence, thereby establishing an optimal routine for task performance. With practice, the execution of routines becomes autonomous, resulting in the acquisition of a new cognitive skill (Gathercole et al., 2019).

Our study aimed to develop a gamified WM intervention that was both engaging and challenging, serving as an enjoyable and effective approach to enhance WM and foster broader cognitive benefits in older adults. The current intervention adopted a gamification design to increase enjoyment and maintain the motivation and persistence of the participants. Crucially, the intervention adhered to the cognitive routine framework, ensuring that the tasks were challenging to encourage new learning and establish novel cognitive routines, thereby enhancing the effectiveness of the training. Specifically, the game required participants to remember and cut specific fruits as they continuously appeared. This task is essentially an updating task (involving the continuous updating of the fruits stored in WM) interpolated with distractors (selection of

specific fruit stored in WM for cutting). To protect stored memory from interference or decay during the presentation of distractors, participants must engage in active rehearsal and exert high-level executive control. This involves encoding (processing incoming fruits), storage, and rehearsal (maintaining activation of recently processed and potentially relevant fruits) within WM, along with inhibitory control processes (discarding and suppressing recently processed but irrelevant fruits; Juvina & Taatgen, 2007). A single process within WM cannot sustain these complex tasks of the current intervention and necessitate coordination with other processes both within and beyond WM (Gathercole et al., 2019). Through practice, participants may develop an optimal and novel cognitive routine to effectively perform the game. This routine involves refining, rearranging, and coordinating the encoding, storage, rehearsal, and retrieval processing of WM and inhibitory control processing. Previous studies have demonstrated that specific factors, such as expectancy, engagement, and motivation, impact the intervention effect (Motter et al., 2016). To control for these factors, the present study deliberately designed a control game that closely resembled the WM intervention game.

It is expected that the establishment of the new cognitive routine and refinement of WM processes during the intervention could yield broad cognitive benefits, given the fundamental role of WM in cognition. The effectiveness of cognitive interventions is often evaluated through near- and far-transfer training effects. Near-transfer refers to improvements in tasks that share overlapping cognitive processes with the directly trained domains, while far-transfer reflects benefits in indirectly trained domains that rely on cognitive processes supported by the trained skill (Turnbull et al., 2022). The digit span backward test, a classic neuropsychological measure of WM (Wechsler, 1981), which involves core WM processes such as encoding, storage, rehearsal, and retrieval (Hilbert et al., 2015; Redick & Lindsey, 2013), was chosen as the near-transfer outcome measure. Episodic memory, inhibitory control, visuospatial processing, processing speed, and attention were chosen as far-transfer outcome measures. These far-transfer effects were assessed using classical neuropsychological tests targeting these domains and were expected to persist for 6 months.

An effective cognitive intervention should not only enhance behavioral performance but also improve neural plasticity in brain regions associated with the trained processes. These neural changes are proposed to be the underlying mechanisms for the observed cognitive enhancements (Duda & Sweet, 2020; Karbach & Schubert, 2013). The prefrontal cortex (PFC) has been suggested to play a crucial role in WM (Friedman & Robbins, 2022). Specifically, the dorsolateral PFC (DLPFC) has been identified as the primary region responsible for WM (Evangelista et al., 2021). Additionally, other regions such as the ventrolateral PFC and the anterior cingulate cortex (ACC) are also thought to be involved (Duma et al., 2019; Lancia et al., 2018).

Resting-state brain function may be particularly informative, as it is more likely to reflect changes in intrinsic brain activity (Reuter-Lorenz & Park, 2014). Although several studies have investigated the effects of WM intervention on resting-state brain function, most of them focused on young adults. After 4 weeks of WM training, resting-state functional connectivity (FC) between the medial PFC (mPFC) and precuneus in young adults increased, while FC between the mPFC and right lateral PFC decreased (Takeuchi et al., 2013). Their subsequent study reported that WM

intervention in young adults led to increases in degree centrality in the mPFC and ACC, increases in fractional amplitude of low-frequency fluctuations in the mPFC, DLPFC, and frontal pole, and increased regional homogeneity (ReHo) spanning from the precuneus to the posterior cingulate cortex and posterior parietal cortex (Takeuchi et al., 2017). Furthermore, following a 3-week *N*-back task intervention, connectivity, measured through independent component analysis, in the right inferior frontal gyrus increased in young adults, while connectivity in the left superior parietal cortex decreased (Salminen et al., 2020). A 10-day verbal WM intervention enhanced resting-state global network modularity in young adults, but decreased in older adults (Iordan et al., 2021).

In this study, by incorporating gamification elements and adhering to the cognitive routine framework, we sought to investigate whether a gamified WM intervention could produce both near- and far-transfer effects in older adults. We hypothesized that the gamified WM intervention could produce significant near-transfer effects on WM performance after the training and would persist for 6 months. Additionally, we postulated that the gamified WM intervention might lead to far-transfer effects on cognitive domains such as episodic memory and inhibitory control, whose underlying processes could be refined during the intervention. At the neurobiological level, we hypothesized that the gamified WM intervention could enhance the structure and function of certain PFC regions, and the neural changes would correlate closely with observed improvements in WM performance.

Method

Participants

The necessary sample size was first estimated through G*Power software (Faul et al., 2007). According to a previous meta-analysis, the effect sizes (Cohen's *d*) of WM interventions on cognition in older adults ranged from 0.35 to 0.50 (Hou et al., 2020). Given two groups and three time points, assuming an effect size of 0.35, a two-tailed test α of .05, a moderate repeated measures correlation coefficient *r* of 0.50, and a statistical power of 80%, a total sample size of 54 was needed. The dropout rate was expected to be 20% (de Vries et al., 2015; Parong et al., 2022), so the target sample size was set at 70.

One hundred twenty-two participants were recruited by posting flyers in local communities, advertising online, and through word of mouth. We enrolled participants who were healthy older adults aged between 60 and 85 years, with more than 6 years of education, normal or corrected-to-normal vision, a score of 24 points or more on the Mini-Mental State Examination, and no video game experience. Participants with psychiatric disorders (e.g., bipolar disorder, depression, schizophrenia) or physical diseases (e.g., cardiovascular disease, neurological disease, disability) were excluded. Participants with claustrophobia or metallic implants (e.g., metal tooth, cardiac pacemaker) in the body could receive the intervention and complete cognitive assessments but were excluded from magnetic resonance imaging (MRI) scans. After the screening, 46 participants were excluded: 36 refused to participate in the intervention due to long-distance commutes, four were excluded due to Mini-Mental State Examination scores of less than 24 points, two were diagnosed with depression, and four had video game experience. Ultimately,

76 participants were included in the present study. All participants received the same amount of compensation throughout the study.

The present study was a randomized, double-blinded study. Participants meeting the inclusion criteria were randomly assigned to the WM intervention group (39 participants) or the active control group (37 participants) through a web application (<https://www.randomizer.org/>). Experimental raters were unaware of the group allocation. Participants in both groups were informed that they were attending a study designed to test the effect of an intervention but did not know that there were two different forms of intervention. Eight participants in the WM intervention group and six in the active control group were excluded from MRI scans due to metal teeth or other metallic implants. One participant in the WM intervention group received the MRI scan but did not complete the cognitive assessments. Therefore, at baseline, 38 participants in the WM intervention group and 37 participants in the active control group completed cognitive assessments. Thirty-one participants in the WM intervention group and 31 participants in the active control group underwent MRI scans.

Data were collected from participants in Beijing, China, from June 2017 to December 2018. The study was approved by the Ethics Committee of the Institute of Psychology, Chinese Academy of Sciences. Written informed consent was obtained from all participants before the study, in accordance with the Declaration of Helsinki.

Experimental Design

All qualified participants underwent MRI scans and completed a battery of cognitive assessments before the intervention. Participants in both groups then participated in twelve 1-hr game training sessions over 6 weeks in the laboratory, involving two sessions per week. After the 6-week intervention period, participants in both groups underwent MRI scans and completed cognitive assessments again. Cognitive function was also assessed at the 6-month follow-up. For each participant, the MRI scans and cognitive assessments were conducted separately.

Gamified Digital Intervention Design

The games that the WM intervention and control groups played involved fruit cutting and were designed according to the principles of game design for older adults (Gerling et al., 2012; Whitlock et al., 2011) in the JavaScript language. The games were presented on a Lenovo computer with a Windows operating system and a screen measuring 345.44 × 259.08 mm; the computer was operated with a mouse. In the game, a fruit cue was always displayed at the top middle of the screen, and different types of fruit ascended from the bottom of the screen. Participants cut the fruit according to the cue. Once the fruit was cut, the fruit cue changed. If the fruit was cut correctly, participants gained 1 point; if not, they lost 1 point. The accumulated game points were continuously displayed in the upper left corner of the screen during the game.

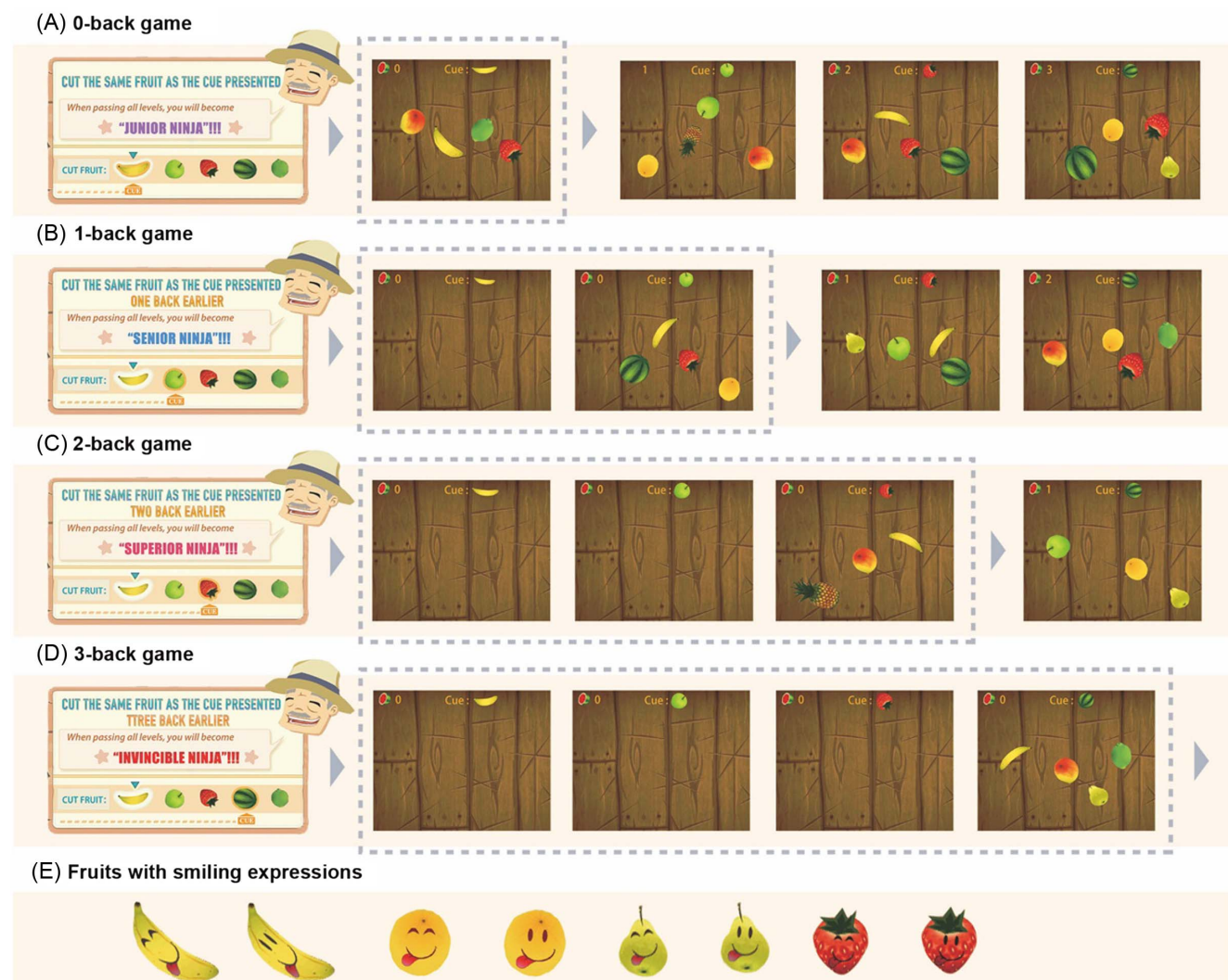
The WM intervention game was designed based on the *N*-back task paradigm and consisted of four stages. The first stage was a zero-back game, requiring participants to cut the same fruit as the current cue (Figure 1A). The second stage was a one-back game, requiring participants to cut the same fruit as the last cue

(Figure 1B). The third stage was adapted from the two-back task, requiring participants to cut the same fruit as the cue presented two cues earlier (Figure 1C). The fourth stage was a three-back game, requiring participants to cut the same fruit as the cue presented three cues earlier (Figure 1D). Each of the four stages includes levels of varying difficulty, allowing the game to adapt to the current abilities of participants. The specific settings of each difficulty level, including the number of distractors and the time of fruit presentation, were determined based on the playing experience and feedback of older adults from the game demo during the game design phase. Specifically, Stages 1–3 each included 10 levels. As the difficulty increases, the number of distractors and the speed of fruit presentation also increase. From Level 1 to Level 3, there is one distractor in each level, and the fruit presentation time starts from 3,800 ms at Level 1 and decreases by 800 ms in subsequent levels. For Level 4 to Level 6, there are two distractors in each level, and the fruit presentation time starts from 3,800 ms at Level 4 and decreases by 800 ms in subsequent levels. In Level 7, there are three distractors, and the fruit presentation time is 5,400 ms. For Levels 8–10, one more distractor is added and the fruit presentation time decreases by 400 ms compared with the previous level. Participants received 1 point if they cut the correct fruit, lost 1 point if they cut the wrong fruit, and received no points if they did not cut the cue fruit. When the score reaches 30, participants can advance to the next level. After completing all levels of a stage, participants will proceed to the next stage. To reduce the frustration caused by Stage 4 (three-back) for older adults, the score requirement per level was reduced to 20 points, and the number of levels was increased to 30. The number of distractors and fruit presentation time in Stage 4 were set in the same way as those in Stages 1–3.

The control game also comprised four stages, all of which were zero-back games without remembering requirements. Participants were instructed to slice fruits based on the current cue displayed at the top center of the screen (Figure 1A). The number of levels in each stage, the number of distractors per level, the duration of fruit presentation per level, and the score required to pass each level were consistent with the settings of the WM game. To ensure that the control game was adaptive, each of the four stages was designed with varying difficulty settings. In Stage 1, the stimuli and distractors consisted of different types of fruit. In Stage 2, fruits featured smiling expressions with closed eyes were introduced as distractors (Figure 1E). In Stage 3, all stimuli and distractors featured smiling expressions with either closed or open eyes (Figure 1E). Stage 4 further increased difficulty by introducing a time limitation for each fruit cut.

Participants were trained in groups of 3–5 people each. A specific account was created for each participant when they first registered for the game. Game performance was recorded in each training session, and participants could continue to play the game at the same level in the next session. During the intervention, the same experimenter was available to ensure participant compliance and address potential problems that might occur during training sessions. To control for expectation and placebo effects, the login interface, game-playing interface, number of game stages, duration of each session, total number of intervention sessions, and training laboratory settings for control games were all identical to the WM intervention group.

Figure 1
Gamified Digital Intervention Design



Note. English instructions and flow charts for the zero-back (A), one-back (B), two-back (C), and three-back (D) games. Examples of fruits with smiling expressions (E). See the online article for the color version of this figure.

Measurements

Game Experience

To assess differences in expectation and experience between the two groups, we investigated the extent of fun, frustration, desire to play the game, and expectations regarding improvements in cognitive function and daily life function using a postintervention 7-point self-rating questionnaire (Kühn & Gallinat, 2014). The scale specifically assessed the level of fun ("Do you think that the game is fun?"), frustration level ("Did you feel frustrated during the game?"), the desire to play the game ("Do you want to continue to play the game?"), expectation of improvements in cognitive function ("Do you think that your memory improved after the intervention?"), and expectation of improvement in daily life function ("Do you think your daily life function improved after the intervention?"). All

participants were asked to complete the scale after 6 weeks of the game intervention. Except for the item on frustration, higher scores on the other four items indicate a more positive experience.

Cognitive Outcome Measures

One near-transfer and five far-transfer cognitive outcome measures were selected to quantify the cognitive benefits of the WM intervention. The digit span backward test (Wechsler, 1981), a classic WM task, served as the near-transfer outcome measure, with the maximum span of digits participants could accurately recall in reverse as the indicator of the task. For the far-transfer outcomes, the Trail Making Test Part A (Armitage, 1946) was used to assess processing speed, with reaction time as the task indicator. The flanker task (Fan et al., 2002) measured inhibitory control, with the

mean reaction time difference between incongruent and congruent trials as the task indicator. The Wechsler Adult Intelligence Scale-Block Design (WAIS-block design) (Wechsler, 1981) evaluated visuospatial processing ability, with the performance score based on accuracy and speed as the task indicator. The number cancellation test assessed attention, with the concentration index as the task indicator (Hatta et al., 2012). Word list recall and word list delayed recall tasks (Andreasen et al., 1995) measured episodic memory, with the final number of words correctly recalled as the task indicator.

Image Acquisition

To investigate structural changes due to the intervention, we collected structural T1 images of participants pre- and post-intervention on a 3T MRI scanner (GE Discovery MR750), interleaved from bottom to top according to the following parameters: repetition time = 6.65 ms, echo time = 2.93 ms, flip angle = 12°, voxel size = $1 \times 1 \times 1 \text{ mm}^3$, slice thickness = 1 mm, slice number = 1 slice, and field of view (FOV) = $256 \times 256 \text{ mm}$.

We also collected resting-state functional MRI (rs-fMRI) images of each participant in a bottom-up interleaved order with the following parameters: repetition time = 2000 ms, echo time = 30 ms, flip angle = 90°, voxel size = $3.5 \times 3.5 \times 3.5 \text{ mm}^3$, slice thickness = 3.5 mm, slice number = 33 slices, and FOV = $224 \times 224 \text{ mm}$.

Data Analyses

Game Performance and Experience

To explore changes in game performance during the intervention, we conducted repeated-measures analyses of variance (ANOVAs) in the WM intervention and control groups. The 12 training sessions served as the independent variable, and the game level reached in each session was the dependent variable.

To test whether there were differences in game experiences between the two groups during the intervention, we conducted independent sample *t*-tests for each item in the self-report questionnaire.

Cognitive Performance

Independent-sample *t*-tests were conducted on the pretest cognitive performance of each test to examine baseline differences between the WM intervention and the active control groups. A 2 (time: pretest, posttest/6-month follow-up test) $\times$ 2 (group: WM intervention group, active control group) repeated-measures ANOVA was conducted to examine the immediate and long-term intervention effects. For post hoc analysis of the within-group changes following a significant main effect in each ANOVA, we performed two-tailed, paired-sample *t* tests on each group separately to test for significant differences from pretest to posttest. Because these tests were performed separately on independent outcome variables, and only one test for each group (WM intervention and active control), we did not perform a statistical correction for multiple comparisons (Ziegler et al., 2019). The effect size (Cohen's *d*; Cohen, 1988) was calculated to determine the intervention effects in the WM intervention group compared to those in the control group by Comprehensive Meta-Analysis software (Version 2.2; Borenstein, 2022). Based on Cohen's *d*, an effect size of less than 0.2 was considered negligible, between 0.2 and 0.5 small,

between 0.5 and 0.8 moderate, and equal to or greater than 0.8 large (Cohen, 1988).

MRI Data Preprocessing and Statistical Analyses

MRI data processing was conducted through the FreeSurfer 7.3.2 image analysis suite (<https://surfer.nmr.mgh.harvard.edu/>). MR images of each time point were first processed cross-sectionally with a standard fully automated cortical reconstruction process stream, including motion correction, transformation, intensity correction, skull stripping, volume labeling, segmentation, surface extraction and modeling, cortical thickness estimation, registration, and parcellation (Fischl, 2012; Fischl et al., 2004; Ségonne et al., 2004). To accommodate the repeated-measurement design, a longitudinal processing stream in FreeSurfer was used to create an unbiased, within-subject template for each participant by robust, inverse consistent registration (Reuter & Fischl, 2011). Cross-sectional images were then registered to this template to ensure a nonbiased analysis (Reuter et al., 2012). The output yielded estimates of gray matter volume, surface area, and cortical thickness for each parcellation on the Desikan-Killiany cortical atlas (Desikan et al., 2006) at each time point.

The present study primarily focused on structural changes in PFC regions. The gray matter volume, surface area, and cortical thickness of 11 PFC parcellations from the Desikan-Killiany cortical atlas in both hemispheres were extracted, including the caudal part of the ACC, the rostral part of the ACC, the caudal part of the middle frontal gyrus, the rostral part of the middle frontal gyrus, the lateral part of the orbitofrontal cortex, the medial part of the orbitofrontal cortex, the opercular part of the inferior frontal gyrus, the orbital part of the inferior frontal gyrus, the triangular part of the inferior frontal gyrus, the superior frontal gyrus, and the frontal pole. Two-way repeated-measures ANOVA was conducted on the changes in gray matter volume, surface area, and cortical thickness in these PFC regions. Bonferroni correction was applied, and $p < .05/11$ was considered significant.

Rs-fMRI Data Preprocessing and Statistical Analyses

The surface-based ReHo (2dReHo) and FC are two widely employed metrics for assessing intrinsic functional interactions and synchronizations between vertices, capturing local and long-range connectivity in resting state, respectively (Zuo & Xing, 2014). Previous studies have demonstrated moderate to high test-retest reliability of these two metrics (Zuo et al., 2013; Zuo & Xing, 2014). In this study, we utilized these metrics to measure PFC functional changes induced by WM intervention.

Rs-fMRI data processing was mainly conducted following the Connectome Computation System pipeline (Xu et al., 2015). The preprocessing steps included: (a) excluding the first five time points; (b) removing and interpolating temporal outliers or spikes due to artifacts from in-scanner head motion or hardware instability; (c) correcting slice timing and head motion across different volumes; (d) normalizing the 4D whole brain mean intensity to 10,000; (e) regressing out nuisance signals of the white matter, cerebrospinal fluid, and the Friston-24 motion curves; (f) performing bandpass filtering (0.01–0.1 Hz); (g) removing both linear and quadratic trends; and (h) matching the spatial correspondence between individual functional and structural images using the boundary-based registration algorithm. The pipeline also computed

the root-mean-square of framewise displacement for in-scanner head motion, excluding participants with a root-mean-square of framewise displacement > 0.5 mm from subsequent analyses.

Each PFC region showing significant changes in the MRI analyses was labeled as a region of interest (ROI). The 2dReHo of each participant at pretest and posttest was calculated within each ROI through Connectome Computation System. 2dReHo is defined as the temporal coherence or synchronization of the blood-oxygen-level-dependent time series among the nearest neighbors of a given vertex. It quantifies the functional interactions or synchronizations between neighboring vertices (Jiang & Zuo, 2016). Briefly, Kendall's coefficient of concordance between the time series of each vertex within each ROI on the fsaverage5 grid and those of its six nearest neighbors, creating vertexwise Kendall's coefficient of concordance-2dReHo values. The average of these vertexwise values was recorded as the final 2dReHo value for the ROI, reflecting the local connectivity on the surface within the ROI. Then, a 2 (time: pretest, posttest) $\times$ 2 (group: WM intervention group, active control group) repeated-measures ANOVA was conducted to examine the changes in 2dReHo caused by the intervention.

The FC of each ROI with the whole brain was also calculated through the Connectome Computation System. Specifically, the mean time series of each ROI was computed and correlated with the pre- and posttest time series of each vertex on the fsaverage5 grid for each participant. All single-subject FC maps were then transferred to Fisher's z FC maps for further spatial smoothing (using a Gaussian kernel with a full width at half maximum of 6 mm) on the fsaverage5 surface grid. In the second-level analyses, a Time $\times$ Group two-way repeated-measures ANOVA was conducted to assess group differences in intervention-related FC changes. Familywise error correction based on 3dClustSim with 1,000 Monte Carlo simulations was used for multiple comparisons, with the vertexwise threshold set at $p < .005$, and the clusterwise threshold set at $p < .05$.

Statistical Learning Analytics

We further investigated the relationship between changes in WM and structural and functional changes of each significant ROI in the PFC. Traditional correlation analysis, which relies on substantial between-subject variability, may be problematic in experimental studies that focus on between-group differences rather than between-subject variability (Hedge et al., 2018). Consequently, we employed statistical learning analytics to evaluate the consistency and discriminative power of intervention-induced changes in WM and the PFC, instead of using correlation analysis.

The average z -scores of pre- and posttest changes in 2dReHo and FC within the ROI were calculated to represent the overall PFC functional changes induced by the intervention. The pre- and posttest changes in significant structural metrics within the ROI represented the PFC structural changes induced by the intervention. The pre- and posttest changes of WM, PFC function, and PFC structure of each participant were used as the three input features. A support vector machine model with a linear kernel was trained for binary classification of the two groups based on these three input features. Leave-one-out crossvalidation was employed to evaluate the classification performance. Classification accuracy was computed by leaving one participant out at a time and predicting the group of this participant based on a training set including all remaining participants. This procedure was repeated for each participant. The

average accuracy from crossvalidations was used as the indicator of the classification performance, representing the discriminative power between the two groups and the consistency of the intervention-induced changes in WM and PFC within each group. The same analysis was also conducted based on pre- and follow-up-test changes in WM, pre- and posttest changes in PFC function, and pre- and posttest changes in PFC structure.

To further validate the classification performance, an additional variable, changes in centile scores of gray matter volume within the ROI before and after the intervention, was introduced as the fourth input feature in the above three-dimensional analysis. The centile scores, derived from lifespan normative models of the human brain, efficiently leveraged priors condensed from population-level neuroimaging data (over 120,000 MRI scans from more than 100,000 participants; Bethlehem et al., 2022). Thus, it is expected that incorporating the centile scores will maintain or even enhance the classification accuracy.

Transparency and Openness

We describe how we determined our sample size, any data exclusions, manipulations, and measures in detail. The study design, hypotheses, and analytic plan were not preregistered. The data, code for data analysis, and related research materials can be accessed on Science Data Bank (<https://cstr.cn/31253.11.sciencedb.23795>; Wang et al., 2025).

Results

Participants

Seventy-six eligible older adults were randomly assigned to either the WM intervention group or the active control group. There were no significant differences between the WM intervention and active control groups in terms of age, sex ratio, years of education, scores of the Mini-Mental State Examination, Multifactorial Memory Questionnaire scores, Center for Epidemiologic Studies Depression scale scores, State-Trait Anxiety Inventory scores, or Activities of Daily Living scores (Table 1).

Table 1
Demographic Information for Participants

Demographic variable	Training group		Control group		t/χ^2	p
	M	SD	M	SD		
Age (year)	67.34	3.31	67.66	6.19	-0.27	.792
Female's ratio	0.66	0.48	0.66	0.48	0.00	1.000
Education (year)	12.80	3.20	12.43	3.37	0.47	.638
MMSE	28.77	1.40	28.66	1.41	0.34	.735
MMQ-contentment	47.40	12.45	45.77	11.02	0.58	.564
MMQ-ability	49.03	12.54	50.03	10.20	-0.37	.715
MMQ-strategy	29.71	14.93	27.91	13.86	0.52	.603
CES-D	14.69	7.77	15.40	7.00	-0.40	.687
S-AI	32.57	9.66	32.74	8.41	0.08	.937
T-AI	35.14	9.20	35.14	10.28	0.00	1.000
ADL	8.11	0.40	8.23	0.73	-0.81	.421

Note. MMSE = Mini-Mental State Examination; MMQ = Multifactorial Memory Questionnaire; CES-D = Center for Epidemiologic Studies Depression scale; S-AI = State-Anxiety Inventory; T-AI = Trait-Anxiety Inventory; ADL = Activities of Daily Living Scale.

The mean dropout rate was 8% at the posttest (10% in the WM intervention group; 5% in the active control group) and 13% at the 6-month follow-up (12% in the WM intervention group; 14% in the active control group), with no significant group differences (posttest: $\chi^2 = 0.61, p = .675$; 6-month follow-up: $\chi^2 = 0.93, p = .999$). Specifically, during the intervention phase, four participants in the WM intervention group and two participants in the active control group dropped out due to illness or unwillingness to continue, resulting in 35 participants (34 behavioral participants, 28 MRI participants) in the WM intervention group and 35 participants (35 behavioral participants, 29 MRI participants) in the active control group. For the included participants, the estimated statistical power of the present study was 89% for the cognitive outcomes and 82% for the neuroimaging outcomes. Thirty-four participants in the WM intervention group (one fewer due to being lost to follow-up) and 32 participants in the active control group (three fewer because they moved to other cities) completed the 6-month follow-up assessment. The estimated statistical power was 88%.

One participant in the WM intervention group did not complete the flanker task in the posttest due to a software malfunction. In the 6-month follow-up test, one participant in the control group was not assessed for the word list delayed recall test because the interval from the word list recall test exceeded 30 min. All other participants completed all cognitive tests in the pretest, posttest, and follow-up test.

Game Performance and Experience

Repeated-measures ANOVA revealed a significant main effect of session across the 12 training sessions in both WM intervention, $F(1, 11) = 198.88, p < .001, \eta_p^2 = 0.85$, and active control, $F(1, 11) = 201.13, p < .001, \eta_p^2 = 0.86$, groups (Figure 2).

The independent-sample *t*-test revealed no significant differences between the WM intervention and active control groups in terms of fun ($t = -0.43, p = .672$), frustration ($t = 0.99, p = .327$), desire to play the game ($t = 3.35, p = .730$), expectations of cognitive

function improvement ($t = -0.16, p = .871$), or expectations of daily life function improvement ($t = -0.40, p = .689$; Table 2). These results demonstrated that both groups had similar game experiences and expectations, suggesting that the active control group served as an effective control group in the present study.

Intervention Effects on Cognitive Performance

The pretest, posttest, and 6-month follow-up performance on cognitive tasks are presented in Tables 3 and 4. There were no significant group differences in any cognitive measures at the pretest (Table 2).

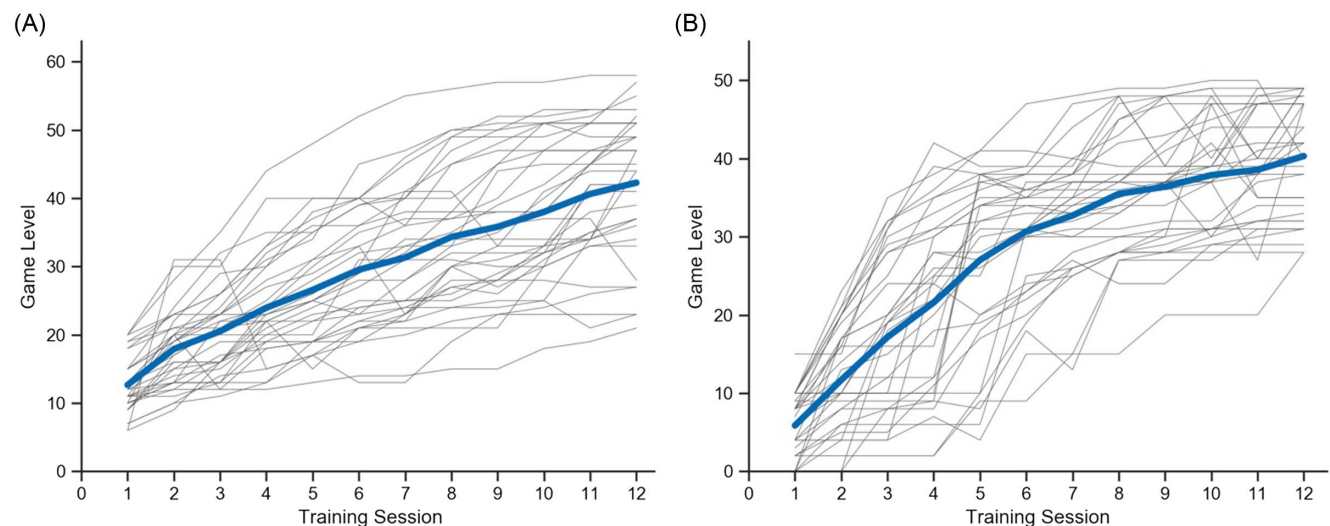
Near-Transfer Cognitive Outcome Measure

Using two-way repeated-measures ANOVA analyses on the digit span backward test, we found that while there was no significant time-by-group interaction at pretest and posttest, $F(1, 67) = 1.73, p = .193, \eta_p^2 = 0.025$, the WM span of the WM intervention group was significantly higher at posttest than at pretest ($t = 2.76, p = .009$). The time-by-group interaction effect was significant at pretest and 6-month follow-up test, $F(1, 64) = 5.40, p = .023, \eta_p^2 = 0.078$, (Figure 3A, Table 4). Post hoc pairwise comparison demonstrated that the WM intervention group showed significant improvement in the digit span backward test from the pretest to follow-up test ($t = 4.38, p < .001$), while participants in the active control group did not show change ($t = 0.85, p = .405$), suggesting a positive effect of the WM intervention on WM. The effect sizes indicated small WM intervention effects on performance in the digit span backward test at posttest ($d = 0.23$) and follow-up test ($d = 0.48$).

Far-Transfer Cognitive Outcome Measures

Two-way repeated-measures ANOVA was conducted separately on the far-transfer cognitive outcome measures at the pretest and posttest. Significant time-by-group interaction effects were found on

Figure 2
Training Curves for the Working Memory Intervention Group (A) and Active Control Group (B)



Note. See the online article for the color version of this figure.

Table 2

Between-Group Differences in Game Experience and Cognitive Performance at Baseline

Variable	Training group		Control group		<i>t</i>	<i>p</i>
	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>		
Game experience						
Fun experience	6.17	1.29	6.29	0.93	-0.43	.672
Frustration experience	2.97	1.69	2.57	1.70	0.99	.327
Desire to play	5.46	1.84	5.31	1.61	0.35	.730
Cognition expectation	5.37	1.63	5.43	1.29	-0.16	.871
Life function expectation	5.34	1.75	5.49	1.17	-0.40	.689
Cognitive performance						
Digit span backward test	4.85	1.42	4.86	1.79	-0.01	.991
Trail making test A	39.70	10.57	43.11	13.86	-1.15	.256
Flanker task	91.76	53.29	75.25	50.54	1.32	.191
WAIS-block design	29.76	7.27	31.43	7.74	-0.92	.361
Number cancellation test	21.12	5.55	20.46	6.38	0.46	.649
Word list recall	6.50	1.81	6.49	1.77	0.03	.974
Word list delayed recall	3.24	2.55	3.71	2.32	-0.82	.417

Note. WAIS = Wechsler Adult Intelligence Scale.

performance in the flanker task, $F(1, 66) = 5.40, p = .023, \eta_p^2 = 0.076$, (Figure 3B), WAIS-block design, $F(1, 67) = 7.52, p = .008, \eta_p^2 = 0.101$, (Figure 3C), and word list delayed recall task, $F(1, 67) = 5.65, p = .020, \eta_p^2 = 0.078$, (Figure 3D, Table 3). No other significant interaction effects were found. Post hoc pairwise comparison demonstrated that compared to pretest, the WM intervention group showed significant improvement on performance in posttest in the flanker task ($t = -4.11, p < .001$), WAIS-block design ($t = 3.53, p = .001$), and word list delayed recall task ($t = 4.74, p < .001$), revealing positive effects of the WM intervention on inhibitory control, visuospatial processing, and episodic memory. The effect sizes revealed moderate WM intervention effects on performance in the flanker task ($d = 0.70$), WAIS-block design ($d = 0.53$), and word list delayed recall task ($d = 0.62$), with a small effect on the word list recall task ($d = 0.26$).

Two-way repeated-measures ANOVA was also conducted for both groups at pretest and 6-month follow-up. Significant time-by-group interactions were found on performance in the word list delayed recall task, $F(1, 63) = 5.92, p = .018, \eta_p^2 = 0.086$, (Figure 3D, Table 4). No other significant interaction effects were found. Post hoc tests revealed that the WM intervention group maintained significantly better performance in the word list delayed recall task ($t = 5.39, p < .001$), revealing long-term intervention effects on episodic memory in the WM intervention group. The effect size of WM intervention maintenance was moderate on the word list delayed recall task ($d = 0.62$), and small on the flanker task ($d = 0.44$) and number cancellation test ($d = 0.22$).

Intervention Effects on the PFC

The two-way repeated-measures ANOVA did not reveal any significant time-by-group interaction effects on gray matter volume or surface area in the PFC. However, for changes in cortical thickness,

two-way repeated-measures ANOVA showed a significant effect of the time-by-group interaction on the cortical thickness of the left frontal pole, $F(1, 54) = 10.12, p = .002, \eta_p^2 = 0.16$. Post hoc pairwise comparison revealed that while no significant changes were found in the active control group ($t = -1.57, p = .129$), the cortical thickness of the left frontal pole in the WM intervention group increased significantly after the intervention ($t = 3.03, p = .006$; Figure 4).

Regarding functional changes, there was a significant effect of the time-by-group interaction on the 2dReHo in the left frontal pole, $F(1, 49) = 6.03, p = .018, \eta_p^2 = 0.11$, (Figure 5A). Post hoc pairwise comparison showed that in the WM intervention group, 2dReHo was significantly decreased after the intervention ($t = -2.90, p = .008$); no significant changes were found in the active control group ($t = 0.47, p = .642$). There was also a significant effect of the time-by-group interaction on FC between the left frontal pole and the left inferior temporal gyrus, $F(1, 49) = 16.24, p < .001, \eta_p^2 = 0.25$, (Figure 5B). Post hoc pairwise comparison showed that FC significantly decreased in the WM intervention group ($t = -2.43, p = .024$) but increased in the active control group ($t = 3.55, p = .001$).

Statistical Learning Analytics of Intervention-Induced Changes

Statistical learning analytics were conducted based on the intervention-induced changes in the WM and the left frontal pole. The results demonstrated a classification accuracy of 78% when considering the changes in both the function and structure of the left frontal pole, along with the pre- and posttest changes of WM (Figure 6A). The classification accuracy was 76% when considering the changes in both the function and structure of the left frontal pole, along with the pre- and follow-up-test changes of WM (Figure 6B). Importing the changes in the centile score of the left frontal pole increased the classification accuracy to 78% and 80%, respectively. All of these classification accuracies exceeded the random threshold of 50%, suggesting the consistency of the intervention-induced changes in both WM and the left frontal pole within each group.

Discussion

This study demonstrated that the current gamified WM intervention enhanced not only WM but also inhibitory control, episodic memory, and visuospatial processing in older adults. Additionally, the intervention resulted in structural and functional alterations in the left frontal pole, located in the most rostral part of the PFC, aligning with the observed improvements in WM.

Gamification Improves the Retention of Working Memory Intervention

In the gamified WM intervention group, the dropout rate of 10% was notably lower than that typically observed in traditional WM interventions, which often exhibit dropout rates ranging from 20% (de Vries et al., 2015; Parong et al., 2022) to over 30% (McAvinue et al., 2013). This finding suggests that gamification may bolster participant engagement and motivation, consequently enhancing the quality of collected data. This aligns with prior research advocating for the integration of gamification into experimental

Table 3
Cognitive Performance in the Pretest and Posttest

Variable	Training group		Paired <i>t</i> (post > pre)	Control group		Paired <i>t</i> (post > pre)	Interaction Group × Time	Effect size (Cohen's <i>d</i>)
	Pre	Post		Pre	Post			
Digit span backward test	4.85 (1.42)	5.47 (1.58)	2.76**	4.86 (1.79)	5.11 (1.68)	1.60	$F(1, 67) = 1.73, p = .193$	0.23
Trail making test A	39.70 (10.57)	33.21 (6.67)	-4.58***	43.11 (13.86)	36.94 (12.05)	-3.75**	$F(1, 67) = 0.02, p = .881$	0.03
Flanker task	95.32 (49.82)	65.60 (32.59)	-4.11***	75.25 (50.54)	70.12 (37.66)	-0.67	$F(1, 66) = 5.40, p = .023^*$	0.70
WAIS-block design	29.76 (7.27)	32.71 (6.13)	3.53**	31.43 (7.74)	31.03 (6.55)	-0.45	$F(1, 67) = 7.52, p = .008^{**}$	0.53
Number cancellation test	21.12 (5.55)	22.64 (5.37)	1.74	20.46 (6.38)	22.17 (5.08)	2.42*	$F(1, 67) = 0.03, p = .870$	0.04
Word list recall	6.50 (1.81)	7.68 (2.36)	3.64**	6.49 (1.77)	7.06 (2.43)	1.45	$F(1, 67) = 1.40, p = .240$	0.26
Word list delayed recall	3.24 (2.55)	5.53 (2.56)	4.74***	3.71 (2.32)	4.31 (2.86)	1.15	$F(1, 67) = 5.65, p = .020^*$	0.62

Note. WAIS = Wechsler Adult Intelligence Scale. Values in parentheses are the standard deviation values.
* $p < .05$. ** $p < .01$. *** $p < .001$.

designs to test benefits or theoretical mechanisms over control conditions (Long et al., 2023; Wang et al., 2023).

The self-report questionnaires revealed no significant group differences in terms of fun, frustration, desire to play the game, or expectations of cognitive improvement or improvements in daily life functioning. The rigorous control over game design ensured that any observed differences between groups could confidently be attributed to the WM intervention, mitigating placebo and expectation effects.

Gamified Working Memory Intervention Produces Near-Transfer Effects

In the gamified WM intervention, participants were required to continuously update information about stored fruits as each new fruit was presented, and to select the correct fruit (stored in WM) to cut amid distractor fruits. According to the cognitive routine framework (Gathercole et al., 2019), these game requirements are complex and challenging for older adults, who struggle to accomplish game tasks using established WM subprocesses. A novel cognitive routine was required to meet the game demands, in which processes of encoding, storage, rehearsal, and retrieval of information in WM and processes of inhibitory control were refined, rearranged, and coordinated. The establishment of the new routine during the WM intervention enabled the improvement of WM and other cognitive functions sharing common processes with WM or the routine (Smid et al., 2020).

Performance on the digit span backward test was significantly improved at the 6-month follow-up in the WM intervention group, indicating that the WM improvements were sustained in older adults after the gamified WM intervention. This finding is consistent with WM benefits reported in previous WM intervention studies (Soveri et al., 2017). The digit span backward test, a complex WM task, involves controlling the rehearsal process and repeated forward retrieval (Thomas et al., 2003). Both processes were refined during the WM intervention, potentially explaining the performance enhancement on this task. Notably, the WM performance enhancement was not immediately evident but was observed 6 months later, contrary to the meta-analysis results of Melby-Lervåg and Hulme (2013), which reported reliable immediate but not sustained WM improvements of WM intervention. Extensive training is necessary for WM intervention efficacy (Klingberg, 2010), with more than 20 training sessions producing significantly larger effect sizes on WM than fewer sessions (Weicker et al., 2016). Given the complex nature of WM, we postulate that improvements in WM processes in the newly established routine during the intervention might take time to manifest behaviorally. The 12 sessions used in this study might not be sufficient to show significant WM improvement immediately afterward.

Previous literature suggests that repeated testing typically leads to improvement due to practice effects (Calamia et al., 2012; Scharfen et al., 2018). In the control group, while not statistically significant, we observed that the control group had improved WM performance at both posttest (mean WM span increased from 4.86 to 5.11) and 6-month follow-up (mean WM span increased from 4.78 to 4.97) compared to baseline, suggesting the presence of practice effects. For the WM intervention group, we speculate that the significant WM improvements observed at the 6-month follow-up test may be

Table 4
Cognitive Performance in the Pretest and 6-Month Follow-Up Test

Variable	Training group		Control group		Paired <i>t</i> (FU > pre)	Pre	Post	Paired <i>t</i> (FU > pre)	Interaction Group × Time	Effect size (Cohen's <i>d</i>)
	Pre	Post	Pre	Post						
Digit span backward test	4.85 (1.42)	5.74 (1.24)	4.78 (1.77)	4.97 (1.66)	4.38***	4.78 (1.77)	4.97 (1.66)	0.85	$F(1, 64) = 5.40, p = .023^*$	0.48
Trail making test A	39.70 (10.57)	33.08 (7.47)	42.20 (14.10)	36.16 (12.36)	-4.61***	42.20 (14.10)	36.16 (12.36)	-3.08**	$F(1, 64) = 0.06, p = .809$	0.06
Flanker task	91.76 (53.29)	61.95 (35.36)	74.32 (50.93)	58.93 (30.70)	-3.85**	74.32 (50.93)	58.93 (30.70)	-1.89	$F(1, 64) = 1.65, p = .204$	0.44
WAIS-block design	29.76 (7.27)	32.88 (7.11)	31.84 (7.51)	33.50 (8.29)	3.84**	31.84 (7.51)	33.50 (8.29)	1.54	$F(1, 64) = 1.20, p = .278$	0.19
Number cancellation test	21.12 (5.55)	23.33 (5.13)	20.58 (6.65)	21.62 (5.41)	2.95**	20.58 (6.65)	21.62 (5.41)	1.36	$F(1, 64) = 1.19, p = .280$	0.22
Word list recall	6.50 (1.81)	7.53 (2.14)	6.50 (1.81)	7.22 (2.80)	2.36*	6.50 (1.81)	7.22 (2.80)	1.55	$F(1, 64) = 0.24, p = .627$	0.13
Word list delayed recall	3.24 (2.55)	5.26 (2.04)	3.77 (2.29)	4.35 (2.62)	5.39***	3.77 (2.29)	4.35 (2.62)	1.24	$F(1, 63) = 5.92, p = .018^*$	0.62

Note. FU = follow-up; WAIS = Wechsler Adult Intelligence Scale. Values in parentheses are the standard deviation values.
* $p < .05$. ** $p < .01$. *** $p < .001$.

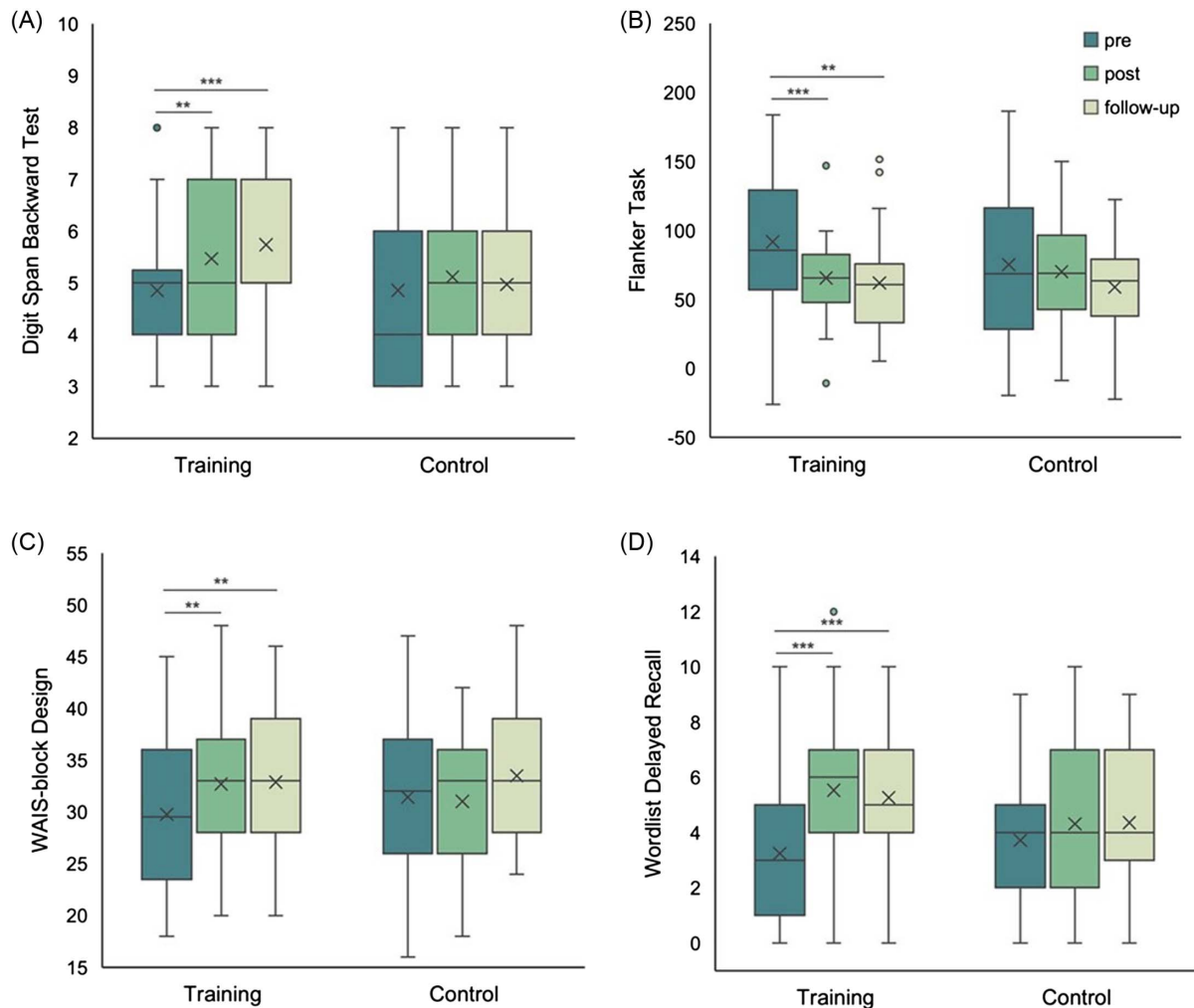
partly attributable to practice effects and partly to the effects of the WM intervention.

Gamified Working Memory Intervention Produces Far-Transfer Effects

In addition to the digit span backward test, the WM intervention group showed significant improvements over the active control group in the word list delayed recall at both posttest and follow-up, and flanker task at posttest. These results demonstrate immediate and sustained improvements in episodic memory and immediate improvement in inhibitory control in older adults due to the gamified WM intervention. WM processes have been suggested to interact with episodic memory (Burgess & Hitch, 2005). Improvement in episodic memory might be due to an enhanced ability to store and maintain information in WM, facilitating the encoding and retrieval of episodic memory (Memel et al., 2019). WM supports inhibitory control by holding goals to determine relevance and what needs to be inhibited. Visual cues that aid people to remember could improve inhibitory performance (Diamond, 2013). Additionally, the WM intervention game included activities involving the suppression of distractors. Therefore, the increased inhibitory control might be due to the refinement of WM processes, the use of fruit cues during the game, and the need to exert inhibitory control while playing the WM intervention game.

The cognitive routine framework suggests that processes within one representational domain cannot be readily adapted to another. The transfer and generalizability of a new cognitive routine developed during WM intervention are limited across representational domains (Gathercole et al., 2019). The current gamified digital WM intervention focused on updating verbal items, so we did not expect a transfer to visuospatial processing. However, while no changes were observed in the control group, increased performance was found in the WM intervention group on the WAIS-block design after training, suggesting that the gamified WM intervention improved visuospatial processing in older adults. We speculate that this improvement might be due to the need to orient to the target fruit to cut it in the WM intervention game, which requires visuospatial processing. This could be refined in the new cognitive routine established during the intervention. The control game had the same requirement but without WM demands, making it easier and less challenging than the WM intervention game. Consequently, it did not enhance visuospatial processing as much as the WM intervention game.

We propose two potential reasons for these observed effects. First, the gamified nature of the intervention may have played an important role. By incorporating gamification elements, the intervention enhanced participants' enjoyment, motivation, and persistence, which are essential prerequisites for effective training (Wang & Li, 2024). Second, and more importantly, the design of the intervention was guided by the cognitive routine framework. This framework posits that new routines help coordinate and integrate cognitive processes, thereby facilitating novel learning and transfer (Gathercole et al., 2019). In contrast to traditional WM training, which typically uses pure WM tasks (Heinzel et al., 2016; Iordan et al., 2021), the present intervention interspersed active updating tasks with distractors, introducing complex demands that cannot be accomplished through preexisting WM processes alone. These demands required the formation of new cognitive routines to reorganize and coordinate WM processes, such as encoding, storage, rehearsal, and retrieval, as well as inhibitory control and

Figure 3*Significant Improvements in Working Memory (A), Inhibitory Control (B), Visuospatial Ability (C), and Episodic Memory (D)*

Note. The “x” in the middle of the box represents the mean value, while the line within the box indicates the median value. The bottom edge of the box corresponds to the first quartile, and the top edge represents the third quartile. The lower whisker denotes the minimum value, and the upper whisker denotes the maximum value. See the online article for the color version of this figure.

** $p < .01$. *** $p < .001$.

visuospatial processing. This may contribute to the transfer effects observed on tasks involving common cognitive elements such as episodic memory, inhibitory control, and visuospatial processing. The significant transfer effects in this study underscore the effectiveness of the cognitive routine framework, suggesting that the framework provides valuable guidance for designing future training programs to achieve broader training effects.

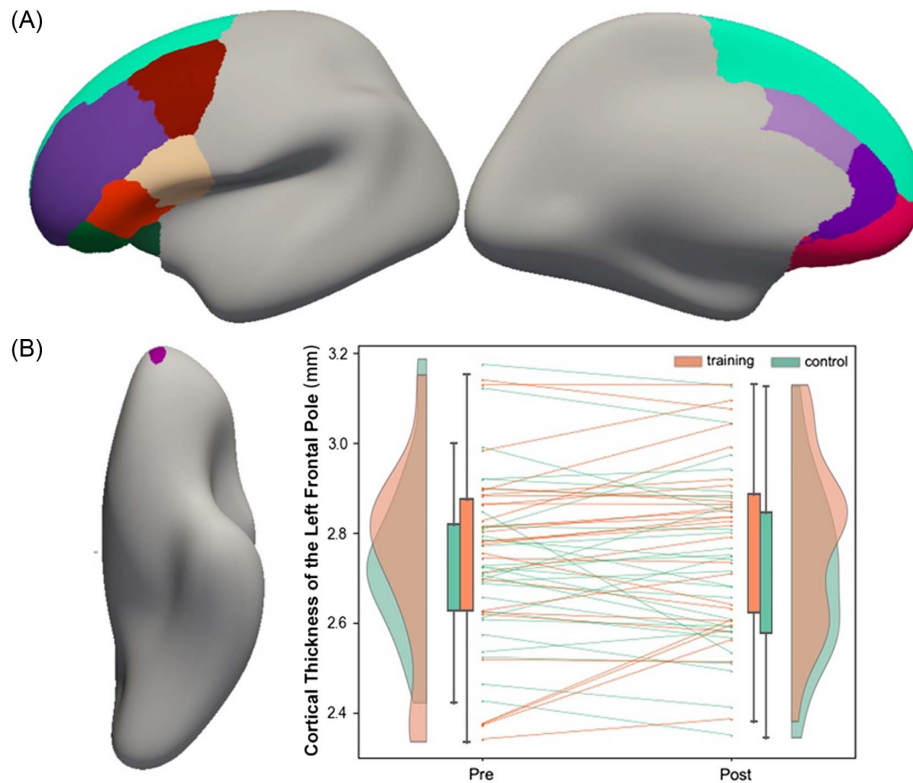
Given that the WM intervention involved inhibitory control (distinguishing targets from distractors) and visuospatial processing (identifying visual characteristics of fruit stimuli), some researchers might consider improvements in these domains to be near-transfer effects. However, it is worth noting that both the WM intervention and the active control game required inhibitory control and visuospatial processing and had similar self-adaptive difficulty levels. If improvements in inhibitory control and visuospatial processing were derived from near-transfer, both groups would have similar

improvements. However, only the WM intervention group demonstrated significant improvements in inhibitory control and visuospatial processing, suggesting that these benefits were not due to practice effects, but rather to far-transfer effects caused by the dynamic integration of WM with inhibitory control and visuospatial processing under increasing cognitive load in the WM intervention group. This integration process facilitates the development of new cognitive routines (Gathercole et al., 2019), which can produce far-transfer effects in untrained tasks. Furthermore, previous studies on WM interventions have also classified measures of inhibitory control and visuospatial processing as indicators of far-transfer (Brum et al., 2020; Juras et al., 2024; Küper & Karbach, 2016).

While our findings demonstrate that gamified WM interventions produce beneficial training effects, it is important to acknowledge that previous studies of gamified interventions have not yielded consistent results (Flak et al., 2019; Nguyen et al., 2022; Nouchi et al., 2012).

Figure 4

Parcellations of the Prefrontal Cortex From the Desikan-Killiany Cortical Atlas (A), and Significant Cortical Thickness Changes in the Left Frontal Pole (B)



Note. See the online article for the color version of this figure.

One possible explanation for these discrepancies is the game experience, which may influence participants' engagement and adherence to intervention. In the present study, participants had a relatively low dropout rate (10%) because they had a good game experience, whereas gamified cognitive training studies that did not show significant training effects tend to have higher dropout rates—often exceeding 15%, and in some cases even exceeding 20% (Finn & McDonald, 2011; Flak et al., 2019; Verghese et al., 2021). Another potential explanation is that the effectiveness of gamification as a tool depends on how effectively the gamification elements and cognitive paradigms are integrated. In the present study, gamification was not simply an ancillary feature but was carefully designed to align with the cognitive routine framework, which emphasizes the formation of new routines to coordinate and integrate cognitive processes (Gathercole et al., 2019). This alignment may enhance the effectiveness of the intervention by ensuring that the gamification elements support the targeted cognitive processes. In contrast, some prior studies have incorporated gamification elements without an overarching cognitive framework, which may have limited their effectiveness. Therefore, the effects of gamified interventions may depend not only on the inclusion of game-like elements but also on the degree to which these elements are integrated with underlying cognitive processes. Future research should focus on designing theory-based gamified interventions and ensure that gamification elements are tailored to support the specific cognitive processes targeted by the training.

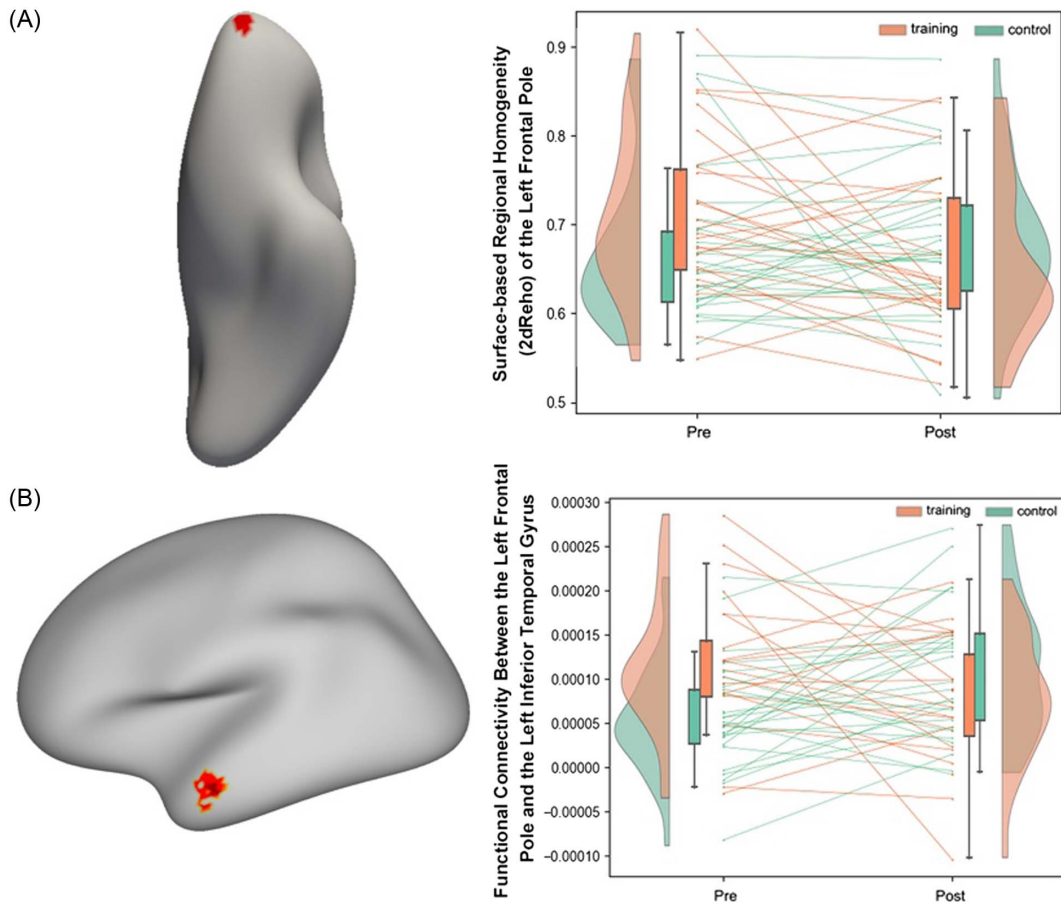
Previous literature on the effects of WM intervention, particularly regarding far-transfer effects, has shown mixed results (Borella et al., 2010; Brum et al., 2020; Gobet & Sala, 2023; Sala & Gobet, 2020). The current results show that gamified WM intervention improved episodic memory, inhibitory control, and visuospatial processing in older adults, but does not improve processing speed and attention, which partially supports the existence of far-transfer effects of WM intervention. These results suggest that the far-transfer effect of WM intervention is not a simple binary outcome but a complex phenomenon. To fully assess and explore the far-transfer effects of WM intervention, researchers are advised to assess multiple cognitive domains rather than limiting themselves to one or two tests, which may lead to biased results. Given the limited prior evidence for the effectiveness of neurocognitive training programs, particularly with respect to far-transfer (Simons et al., 2016), it is important to note that although this study found far-transfer effects at 6-month follow-up, they remain exploratory in nature.

Gamified Working Memory Intervention Improves Plasticity of the Most Rostral Part of the PFC

Previous WM interventions for older adults typically focused on behavioral changes (Booth et al., 2023). While a few studies have investigated the effects of WM intervention on the brain in older adults, most have focused on functional changes (Heinzel et al., 2017; Iordan et al., 2020). By combining both structural and

Figure 5

Surface-Based Regional Homogeneity (2dReHo) Changes in the Left Frontal Pole (A), and Functional Connectivity Changes Between the Left Frontal Pole and the Left Inferior Temporal Gyrus (B)



Note. See the online article for the color version of this figure.

functional changes, the present study could provide a more comprehensive and persuasive picture of the effects of the WM intervention on PFC plasticity in older adults. The findings suggest significant structural and functional changes in the left frontal pole of older adults following the intervention, which is a part of the rostral PFC (rPFC). Specifically, after the intervention, the left frontal pole of older adults exhibited increased cortical thickness, decreased 2dReHo, and reduced FC with the left inferior temporal gyrus. The study of Hedges et al. (2022) has demonstrated that the within-person measurement reliability of cortical thickness exceeded 0.90. Among rs-fMRI metrics, 2dReHo and FC, along with independent component analysis were identified as the most reliable metrics, demonstrating moderate to high levels of reliability (Zuo et al., 2013; Zuo & Xing, 2014). The reliable reliability of the metrics employed in this study ensures the credibility of the obtained results.

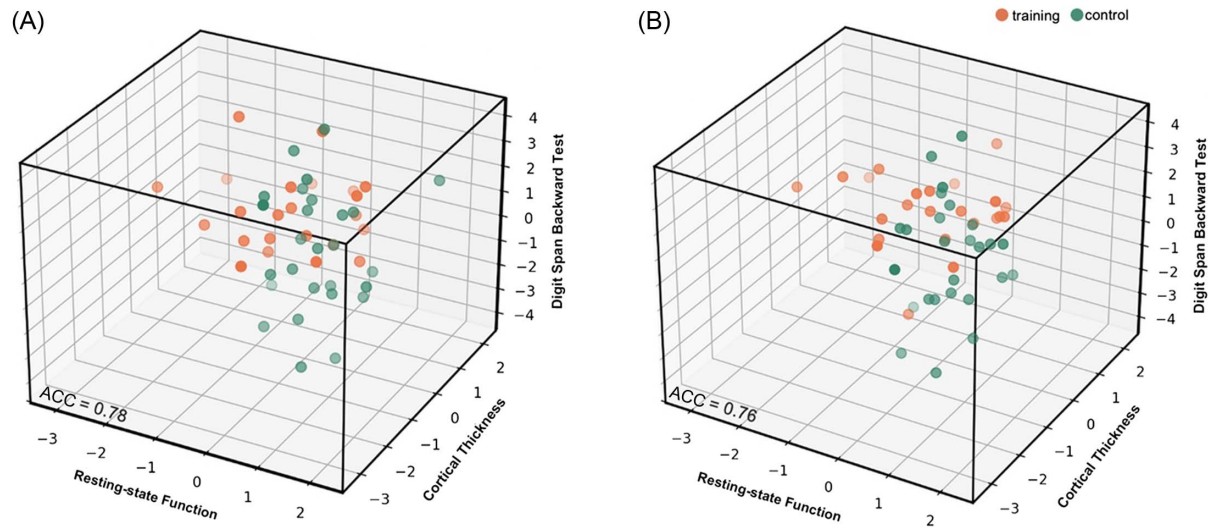
We initially hypothesized that the gamified WM intervention would impact the middle areas of the PFC, particularly the DLPFC, a region primarily responsible for WM (Evangelista et al., 2021) and often influenced by WM interventions in young adults (Miró-Padilla et al., 2020; Nęcka et al., 2021; Salmi et al., 2018). However, the

current gamified WM intervention influenced the rPFC, which is not traditionally considered to be involved in WM (Mansouri et al., 2017). The structural and functional changes observed in the rPFC, rather than the DLPFC, following the gamified WM intervention, may indicate a functional shift from the DLPFC to the rPFC within the PFC with aging. Literature suggests that PFC subregions declined at different rates with age. The rPFC maintains more stable structural integrity throughout the lifespan compared to middle and posterior regions, such as the DLPFC (Bethlehem et al., 2022). This differential decline of PFC subregions may result in a functional shift from the DLPFC to the rPFC with aging. Such a shift aligns with previous studies, which found that while the DLPFC is responsible for the unified representation of executive functions in young adults (He et al., 2021), this role shifted to the rPFC in older adults (Guo et al., 2025). The potential functional shift within the PFC may explain why WM intervention in older adults predominantly influenced the rPFC rather than the DLPFC.

This functional shift within the PFC parallels the shifting cognitive architecture of aging. Spreng and Turner (2019) proposed that cognition undergoes semanticization with aging. When young, individuals rely more on cognitive control and fluid abilities, tending

Figure 6

Statistical Learning Analytics Based on Post-Pre Structural and Functional Changes in the Left Frontal Pole, as Well as Post-Pre (A) or Follow-Up-Pre (B) Performance Changes in Working Memory



Note. ACC = accuracy. See the online article for the color version of this figure.

to focus on external environments and explore novelty (Spreng & Turner, 2019). These novelty-seeking behaviors primarily depend on the middle and posterior regions of the PFC, especially the DLPFC, which is typically considered the top of the PFC hierarchical organization (Badre & Nee, 2018). With increasing age, individuals tend to rely more on internal semantic knowledge, due to the decline in cognitive control resources with an increase in age (Spreng & Turner, 2019). The rPFC, particularly the frontal pole located at the most anterior part of the PFC, characterized by an extremely high density and number of dendritic spines (Hogeveen et al., 2022; Jacobs et al., 2001), has been suggested to be essential for the formation of semantic knowledge and mental schemas (Kroger & Kim, 2022). Its extensive anatomical connections with association regions, especially with the DLPFC, facilitate access to and linking of semantic knowledge and mental schemas (Levy, 2024; Rolls et al., 2024). The cytoarchitecture and anatomical connections of the rPFC, along with its relatively slow decline rates and structural preservation during aging (Bethlehem et al., 2022), may underlie the semanticization of cognition in aging and induce the shift from the DLPFC to the rPFC within the PFC hierarchy.

The PFC is considered the protective region of the whole brain during aging (Davis et al., 2008; Park & Reuter-Lorenz, 2009). If functional shift occurs within the PFC, then the rPFC might act as the protective region of the PFC, or essentially, the protective region of the brain's protective region. Therefore, we propose that the rPFC might serve as a significant biomarker for maintaining normal cognition and daily life function during the aging process. The rPFC could be a crucial target for interventions and treatments in older adults. The PFC undergoes significant structural deterioration in old age (Storsve et al., 2014) and typically shows a trend of increased connectivity or integration (Chan et al., 2014), commonly referring to as functional dedifferentiation and indicating a decrease in neural processing efficiency (Zuo et al., 2017). The present study found that the WM intervention in older adults led to increased cortical

thickness and decreased local and long-range FC in the left frontal pole. These reverse aging changes suggest the plasticity of the rPFC in mitigating the impact of aging on its structure and function through appropriate interventions. It is important to note that this proposed functional shift within the PFC from the DLPFC to the rPFC during aging is speculative. Future studies are needed to further investigate and provide more evidence for this phenomenon.

Possible Influences of the Duration and Frequency of the Intervention

In the present study, the duration and frequency of the intervention (twelve 1-hr sessions over 6 weeks) were determined based on previous studies and practical implementation. First, we drew on an influential study in which they conducted a gamification intervention in older adults, consisting of twelve 1-hr sessions, and found significant near- and far-transfer effects (Anguera et al., 2013). In addition, we reviewed the duration of WM intervention, typically 4–8 weeks (Finc et al., 2020; Heinzel et al., 2016; Jia et al., 2023). Second, the intervention was conducted in a laboratory setting, and we chose the current schedule to try to strike a balance between adequate task exposure and minimizing the risk of participant fatigue and dropout.

The duration and frequency of cognitive intervention are important factors influencing outcomes. The generalizability of the findings may be limited by the specific duration and frequency of the intervention. However, there is currently no consensus on the optimal arrangement of cognitive interventions (Perrot & Maillot, 2023). For example, some studies have suggested that total length and the number of sessions were negatively correlated to training benefits (Teixeira-Santos et al., 2019; Wang et al., 2016), while other studies have found that these factors have no significant effects on training benefits (Owen et al., 2010; Sala et al., 2019). Higher frequency intervention (five or more sessions per week) may reduce

effectiveness in older adults due to increased burden and fatigue (Zhu et al., 2016); however, a recent study suggests that the optimal training schedule for older adults is 50–55 min per day, 6 days a week (Liu et al., 2024). Given these inconsistencies, it remains unclear whether the schedule in the present study is optimal. Future studies are warranted to further explore the effects of training doses.

Limitations and Future Directions

Despite the promising results, several limitations of the present study should be noted. First, a strict active control was used to control for the placebo effect. However, the active control group could also benefit from the intervention, potentially diluting the WM intervention effect. Including both active and passive control groups in future WM intervention studies is recommended. Second, although the WM intervention group showed improvements in WM performance at the 6-month follow-up, this improvement was not observed immediately after training. This delay in improvement needs to be interpreted with caution. Future studies should further investigate whether this result is attributable to the WM intervention, practice effects, or other factors. Finally, the sample size was relatively small, particularly for the neuroimaging analyses. As small data sets may limit the generalizability of findings, the results of this study should be considered preliminary and interpreted with caution. Future studies can use larger sample sizes to validate the current findings.

Conclusion

The current gamified WM intervention significantly improved WM performance and yielded extensive cognitive benefits in older adults, underscoring its potential for preventing cognitive decline in this population. In contrast to the middle areas of the PFC, typically affected by WM interventions in young adults, this gamified WM intervention induced structural and functional changes in the rPFC in older adults, consistent with the improvement of WM. These findings provide initial evidence of a functional shift within the PFC from DLPFC to rPFC and highlight the protective role of the rPFC in brain aging. Further verification and exploration of this phenomenon, along with an in-depth investigation into the underlying neural and cellular mechanisms, are warranted.

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